

ZICHUAN YAO

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Zichuan Yao

Eric Yao

SKILLS

Technical: Python, C, Matlab, Typescript, R, Assembly, VS Code, NextJS, Linux, GitHub, Git, AWS, MS Office

Soft: Bilingual (English and Chinese), Teamwork and Collaboration, Patience, Adaptability, Strong Work Ethic

PROJECTS

Personal Webs | Typescript, HTML, NextJS, Shadcn, Magic UI

- A responsive portfolio website that showcases my work, includes a project gallery, and features contact info.
- [Website address](#)

Iris Neural Network | Python, Numpy, Pandas

- Built a deep neural network from scratch using NumPy to classify Iris flower species. Implemented data preprocessing, He/Xavier initialization, activation functions, forward and backward propagation, and optimization.

WORK EXPERIENCE

Shanghai Jiao Tong University

Sept – Dec. 2025

Research Assistant

Shanghai, China

- Crack Path Analysis Model Optimization & Visualization:** Migrated and refactored a Dijkstra-based crack-path analysis algorithm from MATLAB to Python, added step-wise visualization, and improved documentation for maintainability and clarity.
- Crack Formation Model Development:** Designed a Python model linking crack-initiation density with oxide-film growth in zirconium alloys, converting literature-based mechanisms into an executable simulation tool for quantitative crack-formation studies.
- Technical Documentation & Project Knowledge Base:** Authored clear technical documentation and user manuals for the crack-formation model and optimized analysis program, detailing model principles, function interfaces, and usage workflows to lower the technical barrier for future users.
- Research on Advanced Topics:** Conducted literature reviews on zirconium-alloy corrosion, hydrogen absorption, phase-field modeling, and ML-based materials analysis, providing theoretical support for project direction and experimental design.

Shenglan Semiconductor

Jan – Apr. 2025

Engineer Intern

Shanghai, China

- Semiconductor Process Analysis:** Evaluated bare wafer uniformity by comparing ellipsometry and stress data across process batches, quantifying sample consistency.
- Material Characterization:** Collected and analyzed XRD and SEM data to determine crystal growth orientation and microstructural characteristics of semiconductor materials.
- Experimental Design & Testing:** Tested the piezoelectric effect of cantilever structures and statistically assessed regional uniformity of piezoelectric performance.
- Process Standardization:** Participated in developing a detailed piezoelectric testing operation manual to standardize procedures and parameter settings for consistent, reproducible results.

Porton Pharma Solutions Ltd.

May – Aug. 2024

Data Analyst

Shanghai, China

- Performed multi-source analyses** of global pharma/biotech companies, providing background checks and pipeline transaction insights for partnership decisions.
- Analyzed drug dosage data** across preclinical to commercial stages, benchmarking similar drugs and supporting API-dosage forecasting for unmarketed products.
- Researched the GLP-1 drug landscape** and produced a PowerPoint report outlining key growth drivers and market projections.
- Managed weekly lead data**, ensuring accurate and timely Excel entry for effective lead tracking and follow-up.

EDUCATION

University of Waterloo

Sept. 2023 to May. 2028

Bachelor of Honours Mathematics, Co-operative Education

Waterloo, ON